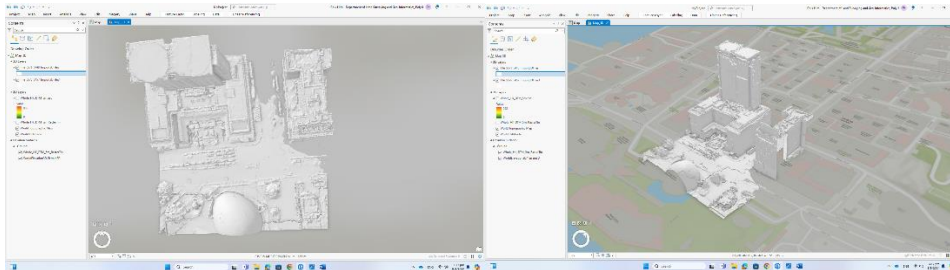


LSGI3242A Digital Terrain Modelling Lab Assignment 1

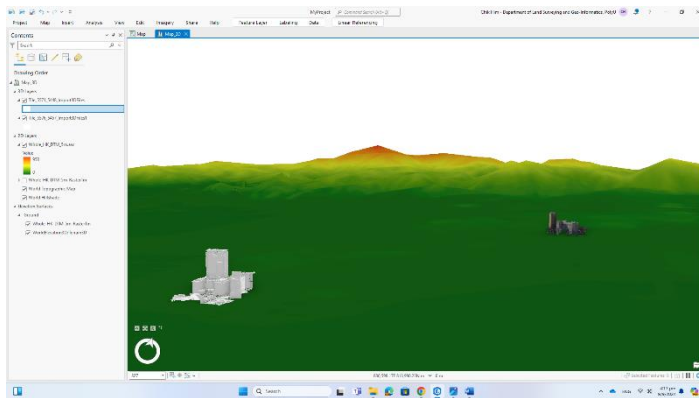
Student name: [REDACTED]

Student ID: [REDACTED]

Task 1 – Integration of 3D mesh model with DTM



(Fig. 1, 2) Final product of the 3D mesh model (11-SW-4D-4) in vertical and isometric view, DTM hidden



(Fig. 3) Final product of the 3D mesh model together with the whole Hong Kong DTM

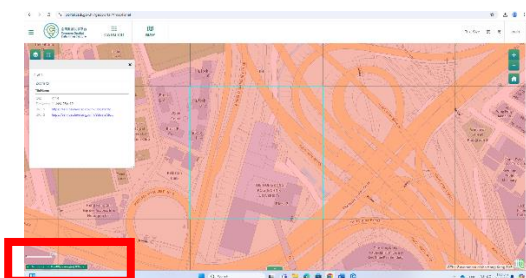
Task 2 – Suggestions for improving the procedures of the integration

When importing the 3D mesh model (model) into the 3D map, 2 problems appear to arise.

The model appears to be located away from Hong Kong under the Hong Kong 1980 Grid coordinate system. This behavior is likely caused by the fact that .obj files do not contain coordinate information. The model also appears to cause lag when dragged around using the Move feature. This behavior is likely caused by the large number of point clouds.

To move the model to the expected location, it's suggested to follow the following procedures with demonstration screenshots:

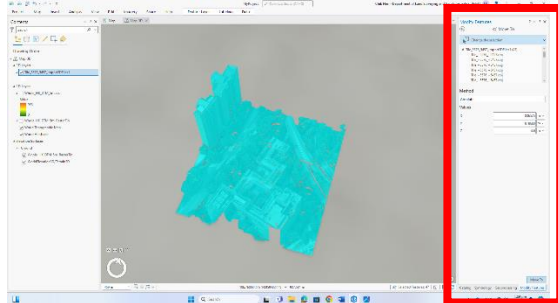
1. find the approximate center coordinate of the 3D mesh model by going to the 3D Visualization Map (2017) on [CSDI](#) > Preview, hover on the grid of the 3D model, then copy the coordinates



(Fig. 4) This demonstration uses 11-NW-25C-19 (PolyU Block Z). Currently the mouse is hovering the approximate center of the square, unfortunately the cursor is not captured

2. locate the model by clicking Zoom to Layer

3. move the model in ArcGIS by clicking Edit > Move To. Paste the coordinates, then click Move To.



4. now the model should match the surrounding features, use the Move tool to make fine adjustments if necessary

The above procedures should also be able to integrate the 3D mesh model. This new procedure pin-points the destination coordinates and uses the Move To feature. Despite this procedure doesn't solve problem with objects placed at default coordinates directly, it does provide a way to move the object approximately to its destination quickly, greatly reducing use of the Move feature, thus reducing the lag caused by the feature, all in all this new procedure should improve the procedures of the integration.

Task 3 - Application of the integration of the 3D mesh model and DTM – Civil Engineering

3D mesh models and digital terrain modelling (DTM) can be used in civil engineering in various ways.

Firstly, 3D models can represent physical objects, including buildings, roads, slopes, waterbodies, and can be used for displaying the features in the designs, increase engagement and serve as visual aid for clients and the general public, as well as for engineers an alternative way to visualize the project.

Secondly, 3D models can be used for feature analysis terrain surface, engineers can analyze the terrain before the excavation and construction works, as well as the structural analysis for the nearby slopes and buildings, evaluating the risks associated with the terrain, thus ensuring the long-term stability and safety. Real time DTM data can also be added to monitor the risks to ensure the safety during the construction and excavation.

In conclusion, the integration of 3D mesh model and DTM in civil engineering can improve the process of visualization and analysis, benefiting both clients and engineers in the project progression and construction safety.